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Meta-analyses on charitable giving clarify evidence for empathic and effective altruism

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Abstract

Effective altruists argue that charitable giving is over-influenced by empathic responses and should instead be guided by cost-benefit analyses about the effectiveness of giving. To provide insight into the extent to which empathy and effectiveness are associated with charitable giving in the population, two meta-analyses are conducted, synthesising 416 effect sizes from 124 papers sampling 74,797 participants. Here we show that effectiveness, $r = .34$, $SE = .04$, 95%CIs [.28, .40], and empathy, $r = .25$, $SE = .02$, CIs [.21, .29], both positively relate to charitable giving overall. However, prediction intervals reveal significant heterogeneity. Moderation analyses reveal one crucial caveat to the overall association between effectiveness and charitable giving: although the mean effect is relatively large when effectiveness is measured, $r = .42$, CIs [.35, .48], manipulated effectiveness has a weak effect, $r = .03$, CIs [-.10, .17]. Our findings suggest that while people may self-perceive as effective altruists, they give like empathic ones, a disjunction that calls for deeper causal investigation.

KEY WORDS: effective altruism; empathy; charitable giving; empathy-altruism hypothesis

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Introduction

Understanding why and to what extent people engage in charitable giving has long been a preoccupation of social scientists.¹ One reason for this is that charitable giving serves an important function in sustaining a healthy and inclusive society. Estimated at over \$470 billion per year in the U.S. alone², charitable giving funds help, care, and services in ways that cannot be readily substituted through commercial or government activities. As such, charitable giving is designed to enrich lives, and in many cases to save lives. Another reason charitable giving has preoccupied scholars is its symbolic value in signifying the degree to which people engage in kindness and compassion to others. Understanding charitable giving helps us understand who we are as humans, and how we express our humanity.

One of the biggest debates about charitable giving – both within and outside academia – is whether people rely too much on their “heart” when giving, and not enough on their “head”. In his book *Against Empathy*³ – and in high-profile opinion pieces⁴ – Paul Bloom makes the case that empathy is a poor moral guide to charitable giving. In line with the empathy-altruism hypothesis^{5,6}, Bloom agrees that empathy plays a role in activating the motivation to do good. However, it also leads us to make ineffective or unfair charitable decisions, focusing on issues that respond to our immediate emotional triggers rather than grounding our compassion in rational deliberation about which causes do most good (i.e., effectiveness considerations). Consistent with this notion, effective altruism scholars propose decision-making frameworks for judging how effective one’s giving will be such as “How many people benefit, and by how much?” and “What are the chances of success?”⁷⁻⁹ Although earlier critiques positioned empathy and effectiveness as competing motives, recent evidence suggests they may be mutually reinforcing, particularly when empathic concern is broadened to include distant others and is guided by reasoning about impact.¹⁰⁻¹³

To assess research on the association between charitable giving, effectiveness, and empathy, a key distinction lies between studies that rely on self-reports of these constructs and studies that experimentally manipulated them. While the debate about the role of empathy in charitable giving has gained increasing attention in recent years, prior syntheses have excluded experimental designs, making this comparison unclear. When it comes to effectiveness, this distinction becomes especially salient given high-profile experiments suggesting that donors have relatively little appetite to seek out information about welfare impact¹⁴, and that when it is provided effectiveness information has limited impact on charitable giving¹⁵ and choice of charities¹⁶.

The goal of this paper is to add an important missing piece to the debate: meta-analyses of the extent to which charitable responses are, in fact, associated with empathy and perceived effectiveness. By combining the voices of thousands of participants sampled by hundreds of researchers across several decades, one is in a stronger position to distinguish signal from noise when examining the associations between charitable giving, empathy, and effectiveness. We emphasise that the goal of this paper is not to compare empathy and effectiveness directly. There are too few studies that measure both constructs within the same sample, and the fields from which these constructs originate use different operationalisations. Instead, our aim is to provide a descriptive and independent synthesis of the extent to which each construct is associated with charitable giving, and under what conditions.

Before doing so, some definitional clarification is required. Empathy is a multi-dimensional construct, one that is often used interchangeably with constructs such as sympathy and compassion. A particularly important distinction is made between “emotional empathy”, which Bloom refers to as “feeling what you believe other people feel – experiencing what they experience” (p.3)³ and “cognitive empathy”, which is variously referred to in the literature as social intelligence, theory of mind, and perspective taking. In *Against Empathy*, Bloom makes it clear that his case is primarily about the limits of affective empathy. So although we meta-analyse all operationalisations of empathy here, we note that

it is affective empathy that is most closely aligned to the debate that inspired the current paper. As illustrated below, empathy has been operationalised in many different ways (e.g., cognitive versus affective; dispositional versus situational; experimentally versus correlationally) and drilling down into the patterns for each helps illuminate the different ways in which empathy may be associated with charitable giving. Empirical synthesis is overdue: the first meta-analysis on the link between empathy and prosocial behaviour was published nearly 40 years ago, and incorporated only a handful of studies on charitable giving.¹⁷ Subsequent syntheses have not examined experimental work, focusing exclusively on correlational links between measures of empathy and questionnaire measures of dispositional pro-social behaviour^{18,19} or between measures of empathy and resource allocations in lab-based games such as the Dictator Game and the Prisoner's Dilemma²⁰.

In extracting data for the effectiveness meta-analysis, we excluded the related literature on efficiency, that is, the proportion of money that goes to beneficiaries as opposed to being absorbed by administrative costs.^{21,22} We made this decision in line with the arguments of effective altruism advocates, who make it clear that the efficiency of a charity should not be a primary criterion for judging whether it is effective (i.e., a charity for which 99.99% of income is funnelled directly towards a pointless mission is efficient but not effective)⁷. Instead, we meta-analyse studies that measure or manipulate variables sometimes referred to as “response efficacy”, “perceived donation efficacy”, or “perceived impact of donation”: constructs that are measured through items such as “How much good do you think you can do by donating money to this organisation?” or “To what extent do you think that your donation would make a positive difference?”

Finally, we restricted our analysis to charitable giving, as distinct from other forms of prosocial behaviour. We defined charitable giving as “voluntary contributions of money that benefit people outside the giver's family”.¹ We had two reasons for restricting the analysis to charitable giving. First, it is an operationalisation of giving that most closely aligns to the effective altruism movement, which is

best known for providing moral context around which charities to support financially. Second, by restricting the range to one domain of prosocial behaviour, we reduce statistical noise and unwanted heterogeneity associated with comparing effect sizes across different behaviours.²³ Patterns of altruism across domains (e.g., donating money versus blood versus time) are psychologically distinct²⁴, so focusing just on charitable giving leaves us in a better position to make like-for-like comparisons between studies.

Here, we show through two meta-analyses that empathy is a robust correlate of charitable giving and reliably increases generosity when experimentally evoked. Perceived effectiveness, on the other, hand predicts giving strongly at the self-report level and weakly under experimental manipulation. We conclude that people give like empathic altruists even when they see themselves as effective ones, a pattern that sharpens the empirical context within which debates about effective altruism unfold.

Results

We followed the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA)^{25,26} to identify quantitative research on (1) the association between effectiveness and charitable giving and (2) the association between empathy and charitable giving. PRISMA flowcharts, summarising the screening process, are presented in Supplementary Figs. S1 and S2. Full reference lists for papers included in the empathy and effectiveness meta-analyses are included in the Supplementary file.

Pearson's correlation (r) was used to represent the association between charitable giving and the two variables of interest: effectiveness and empathy. To approximate a normal sampling distribution, Pearson r correlations were transformed to Fisher's z correlations for analyses. For ease of interpretation, we report values transformed back to Pearson r correlations. Given that the majority of articles included in both meta-analyses reported multiple dependent effect sizes, we employed a three-

level meta-analytic approach to calculate the overall effect size and perform moderator analyses. All tests of significance are 2-tailed unless otherwise specified.

We examined publication bias in two ways: visual inspection of the Funnel Plots (where asymmetry may indicate publication bias) and Egger's Sandwich test, which was used given the presence of dependent effect sizes. As can be seen in Fig. 1, no clear or systematic asymmetry is shown for either meta-analysis. Consistent with visual inspection, Egger's tests were non-significant for both effectiveness ($p = .260$) and empathy ($p = .804$), indicating no statistical evidence of a pattern indicating publication bias.

Heterogeneity and treatment of moderators.

We calculated I^2 (the proportion of total variance in study results due to heterogeneity rather than sampling error) which suggested a high degree of heterogeneity for both the effectiveness (96%) and empathy (92%) meta-analysis. To determine if the within-paper variance (Level 2) and between-paper variance (Level 3) were significant, two one-tailed log-likelihood-ratio tests were performed. These tests compared the fit of the three-level model (which accounted for variance at Levels 2 and 3) to the fit of a two-level model in which variance at only one of the levels was freely estimated. Significant variance in effect sizes was observed at both the within-paper level (both $ps < .001$) and between-paper level (both $ps < .001$). Using Cheung's²⁷ formula, 2.48% of the total variance in the association between effectiveness and charitable giving, and 5.46% of the variance in the association between empathy and charitable giving, was attributed to sampling variance. Considerably more variance is attributable to within-paper variance (32.33% in the effectiveness analysis, 24.11% in the empathy meta-analysis) and between-paper variance (65.19% for effectiveness, 70.43% for empathy). In addition to estimating the overall effect size, we calculated the prediction interval for each meta-analysis. Prediction intervals estimate the likely range of effect sizes that might be observed in future studies with similar characteristics. The range of effect sizes was $r = -.17$ to $r = .71$ for effectiveness and $r = -.14$ and $r = .57$

for empathy. These heterogeneity tests indicate that the true effects vary significantly across papers and measures, and that future studies on these associations may report correlations that vary widely.

Moderation analyses identify likely causes of this heterogeneity. To account for the potential issue of multicollinearity, we followed Hox and colleagues' recommendation to first test for the effect of each potential moderator separately using univariate models.²⁸ We then conducted multiple moderator analyses on a single model that included all significant moderators identified in the univariate analyses for each meta-analysis.

In the main body of this manuscript, we discuss moderators that are most relevant from a theoretical and/or practical perspective. For the sake of thoroughness, however, we also ran ancillary moderation analyses based on sample characteristics: the proportion of women in the sample; the average age of participants; whether data were collected in Western, Educated, Industrialised, Rich, and Democratic (WEIRD) or non-WEIRD countries²⁹; whether the sample comprised students or non-students; and whether the sample was collected using online or offline methods. These analyses are reported in the bottom half of Tables 1 and 2. In short, none of these sampling characteristics were statistically significant moderators of the association between effectiveness and charitable giving (all $ps > .35$) or the association between empathy and charitable giving (all $ps > .08$) and so we do not elaborate on these analyses further.

Effectiveness

Overall, a medium-sized positive association was found between effectiveness and charitable giving, $r = .34$, $p < .001$, $t = 9.85$, 95% CIs[.28, .40]. However, as can be seen in Table 1, three moderators significantly qualified this conclusion. First, the association was significantly stronger when charitable giving was self-reported (e.g., willingness to donate, future intentions to donate, self-reported past giving; $r = .38$, $p < .001$) than when studies used behavioural measures of giving (e.g., direct observations of giving in the field, the amount of a study payment that the participant donated; $r = .27$, p

< .001). Second, the association was significantly stronger when participants reported their likelihood of giving ($r = .37, p < .001$) compared to when participants reported a monetary value of giving ($r = .29, p < .001$). Third, the association between effectiveness and charitable giving was only significant when effectiveness was measured ($r = .42, p < .001$). When effectiveness was manipulated, the average effect was trivial in size and statistically non-significant ($r = .03, p = .612$).

Finally, we examined whether the different operationalisations of effectiveness corresponded to differences in strength of association. In the literature, the dominant operationalisation was what we refer to here as *donation effectiveness*, a construct that is sometimes referred to as “response efficacy”, “perceived donation efficacy”, or “perceived impact of donation”. What unites these operationalisations is that they refer to the extent that the donation will be effective in achieving positive outcomes (example item: “How much good do you think you can do by donating money to this organisation?”). However, as can be seen in Table 1, at least three other operationalisations exist in the literature. What we label as *charity effectiveness* refers to perceptions of the effectiveness of charities in general (measured by items such as “Charitable organisations have been quite successful in helping the needy”). What we label as *collective effectiveness* refers to perceptions of the extent to which collectives can make a difference (e.g., “I think together we can successfully change things in relation to this issue”) whereas *personal effectiveness* refers to perceptions of whether participants can personally make a difference (e.g., “I think it seems possible to make a big difference”). Regardless of how effectiveness was operationalised, a significant and positive association was found, the strength of which did not differ reliably across dimensions.

Following Hox and colleagues’ recommendations²⁸, we extended the three-level meta-regression model to simultaneously include the moderators identified to be significant in the univariate analyses (i.e., measured vs manipulated, self-report vs behaviour, likelihood vs value). The multiple moderator analysis was significant, $F(3,150) = 15.51, p < .001$, revealing a significant unique moderating effect of

measured vs manipulated effectiveness on the overall association between effectiveness and giving (see Table 3 for number of effect sizes and statistical output).

Consistent with the univariate analyses, when effects of the other significant moderators were taken into account, the association between effectiveness and charitable giving was again only significant when effectiveness was measured (and not when effectiveness was manipulated). However, the other moderators (self-report vs behaviour, likelihood vs value) were no longer significant. A possible explanation is that these variables covaried with the measured versus manipulated distinction. Specifically, experimental studies were more likely to employ behavioural outcomes (rather than self-reports) and assess donation value (rather than likelihood), which may have suppressed the independent contributions of these moderators in the full model. In this context, the available data make it difficult to completely disentangle their unique effects within the multivariate analysis.

Empathy

Overall, a medium-sized positive association was found between empathy and charitable giving, $r = .25, p < .001, t = 11.55, 95\% \text{ CIs} [.21, .29]$. Contrary to the pattern of the effectiveness meta-analysis, there was no statistically significant difference in the strength of that association depending on whether empathy was measured or manipulated (both $r_s = .25$, both $p_s < .001$; see Table 2). As in the effectiveness meta-analysis, the association between empathy and charitable giving was significantly stronger when giving was self-reported ($r = .27, p < .001$) than when studies used behavioural measures of giving ($r = .22, p < .001$). The association was also significantly stronger when giving was measured in terms of likelihood of giving ($r = .29, p < .001$) versus the monetary value that participants gave ($r = .21, p < .001$). Moreover, a stronger association was found when empathy was operationalised situationally (i.e., as a state variable; $r = .27, p < .001$) than when it was measured dispositionally (i.e., as a trait variable; $r = .20, p < .001$).

A final moderator was the dimension of empathy that was being captured. A common critique of the field is that researchers' use of the term "empathy" glosses over theoretically meaningful distinctions. In our systematic review, the dominant operationalisation was what we label here as *emotional empathy*, incorporating constructs united by a focus on emotion felt toward a group or individual in need (variously described in the literature as "compassion", "sympathy", "empathic concern", or "affective empathy"). As can be seen in Table 2, the strongest association between empathy and charitable giving emerged when empathy was operationalised this way ($r = .27, p < .001$). Significantly weaker than this (but still significant; $r = .20, p < .001$) were effect sizes in studies in which empathy was operationalised in a more cognitive way (e.g., as a form of perspective taking toward a group or individual in need but without any reference to emotion). Finally, some studies measured empathy as personal distress without explicitly linking that distress to a target group. Most of these were drawn from studies using the personal distress sub-factor of the Interpersonal Reactivity Index, which incorporates items such as "Being in a tense emotional situation scares me".³⁰ When operationalised this way, empathy shared a relatively small association with charitable giving ($r = .09, p = .005$), significantly weaker than for the other two dimensions.

We extended the three-level meta-regression model to simultaneously include the moderators identified to be significant in the univariate analyses (i.e., state vs trait, dimension of empathy, self-report vs behaviour, likelihood vs value). The multiple moderator analysis was significant, $F(5,228) = 20.15, p < .001$, revealing a significant unique moderating effect of state vs trait empathy, dimension of empathy, and likelihood vs value on the overall association between empathy and giving (see Table 4 for number of effect sizes and statistical output).

When effects of the other significant moderators were accounted for, studies that measured state empathy yielded a significantly stronger relation compared to studies that measured trait empathy. Similarly, an identical pattern of findings to the univariate analyses emerged for the dimension of

empathy and likelihood vs value moderators: studies that measured emotional empathy and cognitive empathy yielded a significantly stronger association than those that measured personal distress, and studies that captured giving likelihood yielded a significantly stronger association compared to those that captured giving value. In this model, however, the self-report vs behaviour moderator was no longer significant.

Discussion

We meta-analysed 416 effect sizes from 74,797 participants to answer two fundamental questions: To what extent is empathy associated with charitable giving?; and To what extent is effectiveness associated with charitable giving? Consistent with the empathy-altruism hypothesis^{5,6}, empathy shared a small-to-medium sized association with charitable giving ($r = .25$). This association was statistically robust across a wide range of samples and operationalisations of charitable giving, although significant heterogeneity was observed. The association was strongest when empathy was operationalised in terms of its emotional components, but remained reliable even among studies that measured the cognitive components of empathy or operationalised empathy as a dispositional tendency to experiencing distress, unrelated to a particular target. Importantly, there was no statistically significant difference in the average effect regardless of whether empathy was measured or manipulated. So not only did people's self-reports of empathy predict their charitable giving, but experimenters' invocations of empathy successfully increased charitable giving. In sum, the empirical research indicates that people are indeed influenced by empathy when making decisions about charitable giving.

Although charitable giving is a valid and well-studied expression of altruism, it is only one form among many. It remains an open question whether the patterns we observe here generalise to other forms of altruism that carry higher personal costs, such as organ donation, stem cell registration, or direct physical aid. For instance, Rhoads and colleagues found that self-reported empathic concern was not a consistent predictor of real-world extraordinary altruism in these domains across six populations of

altruistic individuals.³¹ Charitable giving may be more strongly influenced by empathy because it is (a) low-cost, (b) normatively endorsed, and (c) easily enacted. Additionally, some other forms of prosocial behaviour (e.g., volunteering, organ donation, sperm donation) provide more direct payoffs to the “giver” and so may be less empathically motivated.³²⁻³⁴ We encourage future research to test whether the role of empathy in charitable giving is driven by these contextual features, and to explore whether different psychological profiles underlie high-cost versus low-cost altruism.

In our second meta-analysis we found a medium-to-large association between perceived effectiveness and charitable giving ($r = .42$). Despite this, manipulations of effectiveness did not have a statistically significant effect on charitable giving. In short, we identify an apparent “effectiveness paradox”: people self-report in ways that imply effectiveness is very important to them but there was no statistically significant effect of actual effectiveness information presented in experiments.

Why would there be such a discrepancy between the correlational and experimental evidence for the role of effectiveness? One possible answer is that the corpus of experimental studies is simply too small to draw reliable conclusions (this part of the meta-analysis was built on 29 effect sizes, far fewer than available for the correlational analyses). We acknowledge this limitation, although it is possible also that the small number of studies is a consequence rather than a cause of the null result. In the context of a high-profile debate about effective altruism – one that straddles across disciplines within academia but also has extended beyond academia – one might think that experimental tests of effectiveness manipulations would be low-hanging fruit for researchers in this space. The surprisingly small number of studies available to the meta-analysis suggests potential for a file drawer problem: researchers do not seek to progress their null results of publication because they may be (mis)construing them as “failed” experiments that are not diagnostic of real-world behaviours.

With a modest body of work to draw on, it is also possible that the size of the overall association is unreasonably diluted due to subtle manipulations. Speaking against this explanation, however, most

of the effect sizes contained in this section of the meta-analysis were face-valid and relatively blatant. For example, 19 of the 29 effect sizes came from studies in which participants were taught about effective altruism in much the same way as effective altruists themselves present their arguments to the public (e.g., the cost of training a guide dog versus the cost of preventing blindness in developing countries³⁵; comparing the donation response to the “Baby Jessica” incident in Texas with the much larger number of lives that could be saved through other causes⁴). In most of these paradigms, participants are then directed to choose donation amounts for a charity described as being highly effective (versus no information), often using charities that directly correspond to the effective altruism lessons provided in the experimental stimuli. We conducted a posthoc meta-analysis of the effect sizes in which participants were taught about effective altruism. The meta-analysis of just these effect sizes revealed a larger overall effect ($r = .10$) although this was still non-significant, $p = .230$, $t = 1.24$, 95%CIs[-.07, .27]. The fact that blatant manipulations such as these have relatively muted effects on charity decisions – manipulations which explicitly seek to direct participants to more effective choices and willingly invite demand characteristics – suggests a more psychological explanation for the effectiveness paradox.

One such psychological explanation is that people have limited (and imperfect) insight into the drivers of their own behaviour. Decades of research in social psychology have described how self-perception is vulnerable to bias, often skewed in ways that preserve an image of ourselves as both prosocial and logical.³⁶ For example, it seems reasonable that people want to think of themselves as driven by genuinely altruistic motives rather than by the “warm glow” or emotional utility that donations provide. Faced with this self-expectation, it is possible that people rationalise their giving behaviours as effective even when they are primarily driven by less analytical motives (i.e., the causal path flows “backwards” from giving to effectiveness). This could help account for why correlational effects for effectiveness would be larger than experimental effects.

Another possibility is that people see effectiveness as relevant in an abstract sense, but when making specific charitable decisions they de-prioritise effectiveness in a mindful and explicit way. It has been argued previously that many donors see their giving behaviour as a signal of personal choice, and for this reason actively embrace the notion that it should be based on what is personally meaningful to them rather than what might be most effective in the world.¹⁶ In the words of Caviola and colleagues, “For many, a donor choosing a charity is more like a gourmand choosing a restaurant than a doctor prescribing a treatment”.³⁷

Further clues for the effectiveness paradox can be found in the parallel literature on pro-environmental behaviour. That literature reveals strong and consistent evidence that people’s pro-environmental behaviours are positively associated with perceptions that their individual actions can make a difference in terms of mitigating climate change.³⁸ However, there is no consistent evidence that experimenters can increase pro-environmental behaviour by promoting individual efficacy.^{39,40} In other words, the pattern resembles what was uncovered in the current manuscript: strong correlational evidence for the role of effectiveness but non-significant experimental evidence. In making sense of this pattern, some theorists have disputed the classic assumption that ratings of effectiveness flow on from an analytical process: an effortful, mindful, and rational process of weighing up the inputs and outputs of an action. Rather, a case has been made that effectiveness perceptions are formed through non-analytical processes: vulnerable to motivated reasoning, infused with emotion, susceptible to social desirability and identity-expressive concerns, and difficult to change through instruction.⁴¹ If this reasoning is translated into the current context, it might help explain why the correlational evidence for effectiveness as a correlate of charitable giving does not translate into useful interventions. Specifically, self-reports of effectiveness may reflect primarily non-analytic considerations (e.g., emotion, identity, rationalisation) whereas experimental manipulations work from the flawed assumption that people

engage deliberative mental algorithms about effectiveness, processes that can be honed and improved through explicit instruction.

As for any synthesis of existing evidence, limitations of our meta-analysis are a distillation of the limitations of the field. We re-emphasise that meta-analytic syntheses cannot make a definitive statement about whether instruction about effectiveness can have a causal influence on charitable giving. Failure to detect experimental effects does not equate to disproof of causality, and experimental null results can arise from numerous factors (e.g., manipulation weakness, psychological reactance, entrenched priors). Weak experimental effects may also be due to contextual features such as the absence of side-by-side comparisons, variation in donor norms, or limited ecological validity in some lab paradigms. Rather than ruling out an influence of information about effectiveness, the data highlight a tension in the literature: despite robust correlational evidence suggesting that perceived effectiveness predicts giving, experimental attempts to shift perceptions of effectiveness frequently fail to influence donation behaviour. This is a pattern that raises questions rather than one that resolves them, and we call for more nuanced and ecologically valid experimental work to explore potential causal pathways (e.g., longer-term, field-based interventions, or studies that leverage existing variation in perceived effectiveness). Future research should also explore how different dimensions of study quality (e.g., ecological validity, risk of bias, design strength, attrition) may affect conclusions in this area. In sum, our effect sizes should not be treated as literal estimates of variance explained in real-world contexts, but rather as indicators of the extent to which these constructs are currently leveraged successfully in both correlational and experimental work. This distinction highlights the current limits of effectiveness-based interventions and the need for more ecologically valid research in this domain.

Table 1 reveals another clear limitation in the literature: an extreme restriction of range in terms of cultural samples. Strikingly, all but 6 of the 151 effect sizes in the effectiveness library were drawn from so-called WEIRD nations. Going forward, there needs to be much stronger representation from

non-Western samples if the field can get a comprehensive understanding of this global issue. On the upside, over 80% of the effect sizes were drawn from non-student samples, and 32% measured actual donation behaviour as opposed to an intention or some other proxy of behaviour. Although many researchers' choices are presumably constrained by lack of funding, it does suggest a reasonable quorum of evidence drawn from real behaviour among community samples.

Finally, we note that although empathy and effectiveness are often discussed in parallel – both in public discourse and in theoretical debates – there are currently too few studies that measure both constructs within the same design to allow formal statistical comparisons between them. This limits the extent to which we can directly evaluate their relative contributions to charitable giving. We therefore encourage future research to explicitly examine both constructs within the same studies, using consistent measures and parallel designs, to allow more direct comparisons of their predictive value.

In sum, the current analysis sheds light on the current state of evidence regarding two fundamental questions: To what extent is charitable giving associated with empathy? And to what extent is charitable giving associated with effectiveness? The data provide a reliable answer to the first question: empathy is a moderately strong predictor of charitable giving, and interventions that draw on the power of empathy are generally successful. The data provide an intriguing answer to the second question: people self-report to be effective altruists, but there is little evidence that their charitable decisions are, in fact, influenced by effectiveness information. We emphasise that answering these questions do not constitute a “test” of the arguments summarised above about effective altruism and the limitations of empathy: the arguments are complex and not reducible to simple empirical associations between constructs. However, answering these questions will go some of the way to laying a descriptive, empirical context within which philosophical arguments about effective altruism – and practical interventions to improve the impact of charity efforts – can be situated.

Methods

Search strategy

Literature searches were initially run in November 2019 and the search was refreshed in December 2022. The search was conducted across four databases: the two largest multidisciplinary academic databases (*Scopus* and *Web of Science*) plus content-relevant databases in business (*ABI/Inform*) and psychology (*PsycINFO*). For both libraries we used search terms nested around charitable giving (donation*, donate, donor*, philanthrop*, charitable, "not for profit*", "non profit*", nonprofit*, NGO*, "non governmental", "third sector", "charit*") and the factor of interest: effectiveness ("effective altruism", "effective giving", "donation effective*", "charity effective*", "nonprofit effective*", efficacy, competence, "impact of gift", "donation impact", "nonprofit impact", "charity impact"), and empathy (empath*, sympath*, compassion*). Searches were based on title, abstract, and keywords. Results were supplemented through snowballing from reference lists and open calls for unpublished data distributed through X and mailing lists of the *Association for Research on Nonprofit Organizations and Voluntary Action*, *American Marketing Association*, and *Society for Personality and Social Psychology*. To help identify relevant grey literature, we also screened the first 20 pages (200 results) in Google Scholar (consistent with standard practice)⁴² using the same search terms identified in the database searching.

Inclusion criteria and screening

To be eligible for inclusion, studies needed to include a quantitative measure of charitable giving (as part of correlational or experimental designs), be reported in English, and published 1980 or later. Correlational studies allowed us to estimate the strength of naturalistic associations, while experimental studies enabled us to examine causal effects of manipulating empathy or effectiveness on giving behaviour. We excluded purely qualitative, theoretical, or descriptive studies because they do not provide effect sizes or statistical associations necessary for meta-analysis. Similarly, we did not include studies based on simulation models, ethnographic observation, or administrative records without direct

measurement of the psychological constructs of interest, as these designs do not allow for computation of comparable quantitative associations.

In these meta-analyses, we coded for several moderators. In doing so, we defined intention likelihood as a self-reported assessment of the likelihood or willingness to donate to a charitable cause (e.g., “How likely are you to donate to this organisation?”). Value was defined as the monetary value participants reported they would donate or actually donated. We distinguished between measured giving, where donation behaviours or intentions were assessed without experimental manipulation of the constructs of interest, and manipulated giving, where donation behaviours or intentions were measured after an experimental manipulation aimed at influencing empathy or perceived effectiveness. Self-report measures refer to participants’ stated responses to survey items about their donation intentions or behaviour (e.g., likelihood of donating or intended donation amount), whereas behavioural measures involve participants making actual donation decisions, typically involving real or incentivised financial contributions.

There were no limits on age or nationality of participants. Papers that were not full-length articles (e.g., book reviews, books, conference abstracts, corrections, legal judgements) were not included. The second and third authors screened texts for eligibility on title and then again on abstract, resolving discrepancies through discussion. Both screeners had experience conducting and publishing meta-analyses previously. Screening on the full text was conducted by the second author. For articles less than 10 years old that did not report bivariate effect sizes, we emailed authors to request this information.

Analytic strategy

Following the screening process, data on the association between effectiveness and charitable giving were analysed from 47 papers, comprising 81 independent samples and 157 effect sizes ($N = 32,116$). Data on the association between empathy and charitable giving were analysed from 77 papers,

comprising 105 independent samples and 259 effect sizes ($N = 42,681$). Where an experiment both manipulated and measured effectiveness (for example as a manipulation check) both effect sizes were included (one in the “manipulated” category and one in the “measured” category).

We employed a three-level meta-analytic approach to calculate the overall effect size and perform moderator analyses. To determine the appropriate level at which to model statistical dependence among effect sizes, we compared alternative multilevel random-effects structures and modelled dependence at the paper level; model comparisons and sensitivity analyses are reported in the OSF supplement. The total variance is partitioned into three components: sampling variance (Level 1), within-paper variance reflecting dependence among studies in the same article (Level 2), and between-paper variance (Level 3), thereby appropriately accounting for the clustering of effect sizes. To conduct our analyses, we followed the procedures outlined by Assink and Wibbelink⁴³ and used the `rma.mv` function of the Metafor package⁴⁴ in the R environment (version 4.4.0)⁴⁵. We used a multilevel random effects model, and restricted maximum likelihood estimation method to calculate all model parameters. The t -distribution was used to test individual regression coefficients and for calculating confidence intervals⁴⁶.

The type of bivariate effect size extracted depended on study design, but we typically extracted Pearson’s correlation coefficients (r) for correlational studies or Cohen’s d s for experimental studies. In situations where these values were not provided by the study, we used test statistics (e.g., means and standard deviations, F -values, frequency data) to calculate effect sizes. All effect sizes were transformed to Pearson r correlations for the meta-analysis (see OSF for effect size conversion formulas).

A leave-one-out analysis did not identify any studies that influenced the overall findings. To check for extreme effect sizes that may disproportionately influence the results, we screened for those with standardised scores larger than 3.29 or smaller than -3.29.⁴⁷ No outliers were identified in the effectiveness meta-analysis, and only one outlier was identified in the empathy meta-analysis.

Regardless of whether this effect size was included or excluded, analyses yielded similar findings, so we opted to include this effect size in all subsequent analyses.

Data availability statement

The data are publicly available on the Open Science Framework:

<https://doi.org/10.17605/OSF.IO/QZ9AX>.⁴⁸

Code availability statement

Analyses were conducted through R, using code that is viewable on the Open Science Framework:

<https://doi.org/10.17605/OSF.IO/QZ9AX>.⁴⁸

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Author contributions:

C.C. and M.H. acquired funding and conceived the study. C.C. supervised the project and led data coding. J.S. conducted the data extraction, formal analyses, and visualisation. M.H. wrote the first draft of the manuscript. All authors contributed to reviewing and editing subsequent drafts.

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Table 1*Moderators of the Overall Association between Effectiveness and Charitable Giving*

Moderator variable	<i>k</i>	Moderation omnibus test	<i>r</i> / β	95% CIs	<i>t</i> ₀	<i>p</i>
Measured v/ Manipulated	157	<i>F</i>(1,155)=32.22, <i>p</i><.001				
Measured	128		.42	.35, .48	11.61	<.001
Manipulated	29		.03	-.10, .16	0.51	.612
Effectiveness Type	157	<i>F</i>(3,153)=1.04, <i>p</i>=.379				
Donation effectiveness	85		.35	.28, .41	9.41	<.001
Charity effectiveness	30		.30	.18, .40	5.12	<.001
Collective effectiveness	9		.25	.08, .41	2.89	.004
Personal effectiveness	33		.36	.26, .45	7.04	<.001
Self-report v/ Behaviour	157	<i>F</i>(1,155)=7.48, <i>p</i>=.007				
Self-report	106		.38	.32, .44	10.77	<.001
Behavioural	51		.27	.19, .35	6.34	<.001
Likelihood v/ Value	155	<i>F</i>(1,152)=6.20, <i>p</i>=.014				
Likelihood	106		.37	.30, .43	10.34	<.001
Value	48		.29	.21, .36	7.09	<.001
% Women	147	<i>F</i>(1,145)=0.89, <i>p</i>=.348	-.00	-.01, -.00	--	.348
Mean Age	115	<i>F</i>(1,113)=0.13, <i>p</i>=.723	-.00	-.01, .01	--	.723
Pub. status	157	<i>F</i>(1,155)=4.40, <i>p</i>=.038				
Published	148		.36	.30, .42	10.31	<.001
Unpublished	9		.14	-.08, .34	1.27	.207
Sample nation	151	<i>F</i>(1,149)=0.27, <i>p</i>=.603				
Non-WEIRD	6		.40	.17, .59	3.31	.001
WEIRD	145		.34	.28, .41	9.49	<.001
Online v/ offline	157	<i>F</i>(1,155)=0.35, <i>p</i>=.553				
Offline	35		.31	.18, .43	4.64	<.001
Online	122		.35	.28, .42	8.86	<.001
Student v/ other	157	<i>F</i>(1,155)=0.09, <i>p</i>=.769				
Non-student	131		.34	.27, .40	9.13	<.001
Student	26		.36	.22, .48	5.03	<.001

Notes. For continuous moderators, we report β (i.e., estimated regression coefficient) which conveys the increase in mean *r* (mean effect size denoting the association between effectiveness and charitable giving) for a unit increase in a given moderator variable. For categorical moderators, we report the mean *r* for each level of a given moderator variable. For each category of moderator variable, the reference variable is listed first. Text in bold are moderation tests.

k = number of effect sizes contributing to analysis.

t_0 = difference in mean r with zero

Table 2

Moderators of the Overall Association between Empathy and Charitable Giving

Moderator variable	k	Moderation omnibus test	r/β	95% CIs	t_0	p
Measured v/ Manipulated	259	$F(1,257)=0.01, p=.910$				
Measured	245		.25	.21, .29	11.40	<.001
Manipulated	14		.25	.15, .35	4.66	<.001
State v/ Trait Empathy	256	$F(1,254)=7.06, p=.008$				
State empathy	162		.27	.23, .31	12.07	<.001
Trait empathy	94		.20	.15, .25	7.55	<.001
Dimension of Empathy	234	$F(2,231)=26.12, p<.001$				
Cognitive empathy	51		.20	.15, .25	7.49	<.001
Emotional empathy	156		.27	.22, .30	12.52	<.001
Personal distress	27		.09	.03, .15	2.86	.005
Self-report v/ Behaviour	259	$F(1,257)=3.95, p=.048$				
Self-report	155		.27	.22, .31	11.59	<.001
Behavioural	104		.22	.17, .27	8.22	<.001
Likelihood v/ Value	254	$F(1,252)=12.08, p=.001$				
Likelihood	124		.29	.24, .33	12.05	<.001
Value	130		.21	.17, .26	8.80	<.001
% Women	223	$F(1,221)=1.30, p=.256$	.00	-.00, .00	--	.256
Mean Age	174	$F(1,172)=1.04, p=.309$	.00	-.01, .00	--	.309
Pub. status	259	--				
Published	257					
Unpublished	2					
Sample nation	243	$F(1,241)=0.62, p=.430$				
Non-WEIRD	30		.21	.10, .31	3.80	<.001
WEIRD	213		.25	.20, .30	10.31	<.001
Online v/ offline	245	$F(1,243)=2.93, p=.088$				
Offline	175		.23	.18, .28	8.60	<.001
Online	70		.29	.23, .35	9.27	<.001
Student v/ other	259	$F(1,257)=1.00, p=.317$				
Non-student	170		.24	.19, .28	9.75	<.001
Student	89		.28	.21, .34	7.57	<.001

Notes. For continuous moderators, we report β (i.e., estimated regression coefficient) which conveys the increase in mean r (mean effect size denoting the association between empathy and charitable giving) for a unit increase in a given moderator variable. For categorical moderators, we report the mean r for each level of a given moderator variable. For each category of moderator variable, the reference variable is listed first. Text in bold are moderation tests.

k = number of effect sizes contributing to analysis.

t_0 = difference in mean r with zero

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Table 3

Statistical Output from the Multiple Moderator Analysis of the Association between Effectiveness and Charitable Giving

Moderator	Reference Category	β	95% CI		<i>t</i>	<i>p</i>	Notes. For all
			Lower	Upper			
Measured v/ Manipulated	Measured						
Manipulated		-.39	-.50	-.27	-6.19	<.001	
Self-report v/ Behaviour	Self-report						
Behavioural		-.05	-.16	.05	-1.04	.299	
Likelihood v/ Value	Likelihood						
Value		-.04	-.13	.05	-0.80	.425	

moderators (in bold), we report β (the regression coefficient), which reflects the difference in mean *r* (the mean effect size denoting the association between effectiveness and charitable giving) between the comparison category and the reference category. We also report 95% confidence intervals for these regression coefficients. All moderators are dummy-coded, with the reference category noted; *t* = significance of the moderator in relation to the reference category. This analysis was based on 154 effect sizes.

Table 4

Statistical Output from the Multiple Moderator Analysis of the Association between Empathy and Charitable Giving

Moderator	Reference Category	β	95% CI		<i>t</i>	<i>p</i>	Notes. For all
			Lower	Upper			
State v/ Trait	Trait						
State		.08	.04	.12	3.65	<.001	
Dimension of Empathy	Personal Distress						
Cognitive Empathy		.10	.05	.15	3.74	<.001	
Emotional Empathy		.18	.13	.22	7.35	<.001	
Self-report v/ Behaviour	Self-report						
Behavioural		.02	-.03	.07	0.73	.465	
Likelihood v/ Value	Likelihood						
Value		-.11	-.15	-.07	-5.02	<.001	

moderators (in bold), we report β (the regression coefficient), which reflects the difference in mean *r* (the mean effect size denoting the association between empathy and charitable giving) between the comparison category and the reference category. We also report 95% confidence intervals for these regression coefficients. All moderators are dummy-coded, with the reference category noted; *t* = significance of the moderator in relation to the reference category. This analysis was based on 234 effect sizes.

Figure Captions

Fig. 1. Tests of publication bias. Funnel plot plotting *Fisher's z* against the standard error from the effect sizes included in the meta-analyses on effectiveness (top panel) and empathy (bottom panel).

Editor Summary

There is debate about whether charitable giving is driven more by empathy or by reasoning about effectiveness. Here, the authors show via meta-analysis that both correlate with giving, but only empathy reliably boosts giving when experimentally increased.

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